Dean Cruz

+65 9794 0420 | deancruzgg@gmail.com | deancruz1.github.io | github.com/deancruz1 | linkedin.com/in/dean-cruz

Summary

Software Engineer at NCS (AI Engineering & AgentOps) and part-time UOW Computer Science (AI and Big Data) student. Delivers enterprise Java modernisation using AI-assisted, spec-driven workflows, with full-stack experience across Angular, Spring Boot, React, and TypeScript. IMDA Gold Medalist and top RP graduate.

Experience

AI Software Engineer 2026 – Present
NCS Group *Singapore*

- Migrated 800+ raw JDBC queries to MyBatis 3, then reviewed and optimised the full set of 1,600+ queries to fetch only required data, improving efficiency and maintainability with no behavioural regressions
- Owned the JasperReports 6 → 7 upgrade on a legacy enterprise application, fixing a production bug where reports rendered incorrectly in Excel exports
- Migrated 6 EJB endpoints to SOAP web services (provider and consumer), with 24 more in progress
- Built a full-stack Angular and Spring Boot application, designing key front-end components and implementing all backend functionality to specification
- Use NCS's in-house AI agent and OpenSpec (spec-driven development) to plan, implement, and review changes across legacy Java codebases

Software Test Engineer (Intern) 2024 – 2025
Oversea-Chinese Banking Corporation *Singapore*

- Automated 800+ regression and UAT test cases in Tosca and executed 300+ manual test cases in two weeks
- Presented feasibility studies on AI-driven test automation tools to senior stakeholders, influencing team tooling decisions

Projects

Kiroku | *Next.js, TypeScript, PostgreSQL, Tailwind CSS, Prisma*

- Full-stack anime tracker with Discord OAuth, email/password authentication, and JWT session management
- 7-day seasonal broadcast calendar with episode countdown timers and Framer Motion animations
- Genre-based recommendation engine built from watch history, plus a Recharts stats dashboard, using the Jikan API with exponential backoff for rate limits

Urara | *React, TypeScript, Tailwind CSS, Framer Motion, TanStack Query*

- Fan encyclopedia for Umamusume Pretty Derby using the umapyoi.net API, with Japanese-to-English translation via the MyMemory API
- Character browser with animated detail panels and a music player with character filtering and Apple Music previews
- Added a birthday calendar, searchable news feed, and dark/light theming, with responsive layouts across mobile, tablet and desktop

SkillForge | *Angular, Spring Boot, Hibernate, PostgreSQL*

- Full-stack e-learning platform with instructor and student roles, quiz creation, and progress tracking
- RESTful APIs with Spring Boot and Hibernate ORM, backed by a Supabase PostgreSQL database
- Secured 20+ endpoints with Spring Security (JWT, BCrypt, role- and owner-based access); backend on Render via Docker, frontend on Vercel

Education

University of Wollongong 2026 – 2028
Bachelor's in Computer Science (Artificial Intelligence and Big Data), Part-time *Singapore*

Republic Polytechnic 2022 – 2025
Diploma in Digital Design and Development *Singapore*

- GPA: 3.97 / 4.00 | IMDA Gold Medal (2025) | 2x Diploma Prize (AY2022, AY2023)

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, PHP, C#
Backend & Data: Spring Boot, MyBatis, Hibernate, JDBC, REST APIs, JasperReports, PostgreSQL, MySQL
Frontend: Angular, React, Next.js, Tailwind CSS
Tools & AI: Git, Tosca, Claude Code, AI coding agents, OpenSpec, Vercel, Render